=== src/m3/paper.html === <section> Real Analysis II (MAT5CJ301) · Semester 5 · Module 3 · Model paper <h1>The Riemann Integral</h1> Bartle & Sherbert §7.1 (7.1.1–7.1.7), §7.2 (7.2.1–7.2.13), §7.3 (7.3.1–7.3.18) · FYUGP pattern (April 2026), doubled How this paper works A real FYUGP paper has 10 short (3 marks, ceiling 24), 8 paragraph (6 marks, ceiling 36) and 2 essay (10 marks, answer one) questions. This paper has twice as many (20 + 16 + 4 = 40 questions, 196 marks), all from Module 3. Each of the module's 32 atomic concepts (T01–T32) and 8 core application types is the main point of exactly one question. Time4 hours (or two sittings of 2 hours) Maximum140 marks (A ceiling 48, B ceiling 72, C answer any two = 20) PYQ MTS-25 Q6 = MTS 6B 10, March 2025, question 6 (old CBCSS course, same textbook). "also" = another paper asked the same thing. "adapted" = a 1-mark MCQ turned into a 3-mark or 6-mark question with full justification. </section>
Section A · Short answer<small>All questions can be answered. Each carries 3 marks. Ceiling 48.</small>
1. Define a partition of [a,b] and the norm of a partition. Find the norms of P_1 = (0,1,2,4), P_2 = (0,2,3,4), and P_3 = (0,1,1.5,2,3.4,4) on [0,4]. Which of Q_1 = (2,4,6,8,9), Q_2 = (2,6,9), Q_3 = (2,5,7,9) on [2,9] has the largest norm? PYQ MTS-26 Q4 also MAT-22 Q3, MAT-21 Q5 MAT-22 MCQ 16 · adapted
2. Define a tagged partition and the Riemann sum S(f;\dot P) of a function f : [a,b] \to \mathbb{R}. For f(x) = x^2 on [0,4] and P = (0,1,2,4), compute S(f;\dot P) with tags at left endpoints and at right endpoints. PYQ MTS-22 Q4
3. Let f : [a,b] \to \mathbb{R}. State the \varepsilon–\delta_\varepsilon definition of "f is Riemann integrable on [a,b]" (including the notation \mathcal{R}[a,b] and \int_a^b f). Why is \int_a^b f not an ordinary function limit as \lVert P\rVert \to 0? PYQ MAT-21 MCQ 9 · adapted
4. Prove that if f \in \mathcal{R}[a,b], then the value of the integral \int_a^b f is uniquely determined.
5. Prove directly from Definition 7.1.1 that every constant function f(x) = k on [a,b] is Riemann integrable on [a,b], and find \int_a^b k. PYQ MTS-25 Q4
6. If g \in \mathcal{R}[a,b] and f(x) = g(x) except for a finite number of points in [a,b], prove that f \in \mathcal{R}[a,b] and \int_a^b f = \int_a^b g (proving the case of one exceptional point suffices). Hence find \int_0^2 h where h(x) = 2 on [0,1), h(1) = 3, and h(x) = 1 on (1,2]. What happens if f(x) = 0 except at finitely many points, or if f(x) = g(x) off the infinite set \{1/n : n \in \mathbb{N}\}? PYQ MAT-22 MCQ 18 · adapted also MAT-21 MCQ 13, MTS-22 MCQ 7 · adapted
7. State and prove the Boundedness Theorem for Riemann integrable functions. Deduce that H(1/k) = k (k \in \mathbb{N}), H(x) = 0 elsewhere on [0,1], is not in \mathcal{R}[0,1]. Does f \in \mathcal{R}[a,b] imply f is continuous or monotone? Is every bounded function integrable? PYQ MAT-22 MCQ 15 · adapted also MAT-21 MCQ 11, 17, MTS-22 MCQ 8, 9 · adapted

8. If J is a subinterval of [a,b] with endpoints c < d and \varphi_J(x) = 1 on J, 0 elsewhere on [a,b], prove that \varphi_J \in \mathcal{R}[a,b] and \int_a^b \varphi_J = d - c. Hence evaluate \int_0^4 g where g(x) = 3 for 0 \le x \le 2 and g(x) = 2 for 2 < x \le 4. PYQ MAT-21 MCQ 10 · adapted
9. For f \in \mathcal{R}[a,b] and \alpha < \beta in [a,b], define \int_\beta^\alpha f and \int_\alpha^\alpha f. State the additivity formula for arbitrary \alpha, \beta, \gamma \in [a,b] and verify it for f(x) = x with \alpha = 3, \beta = 1, \gamma = 0.
10. Explain why the First Form of the Fundamental Theorem of Calculus (Theorem 7.3.1) cannot be applied to H(x) = 2\sqrt x on [0,b], nor to K(x) = x^2\cos(1/x^2) (x \in (0,1]), K(0) = 0, on [0,1], even though H is continuous on [0,b] and K is differentiable at every point of [0,1].
11. Define the indefinite integral F of f \in \mathcal{R}[a,b] with basepoint a. Prove that if \lvert f(x)\rvert \le M on [a,b], then \lvert F(z) - F(w)\rvert \le M\lvert z - w\rvert for all z, w \in [a,b], and deduce that F is continuous on [a,b]. PYQ MAT-21 Q6
12. Find the indefinite integral F(x) = \int_{-1}^x \operatorname{sgn} of f(x) = \operatorname{sgn} x on [-1,1] with basepoint -1. Is F an antiderivative of f on [-1,1]? What about H(x) = \int_0^x h for Thomae's function h on [0,1]?
13. Define a null set in \mathbb{R}. Prove that the set \mathbb{Q} \cap [0,1] of rational numbers in [0,1] is a null set. State whether subsets and unions of null sets are null, and whether every null set is countable. PYQ MTS-25 Q7 also MTS-26 Q7 MTS-22 MCQ 10 · adapted
14. Using the First Form of the Fundamental Theorem of Calculus: (a) state \int_a^b f(x)\,dx when f is continuous and F is an antiderivative of f on [a,b]; (b) evaluate \int_{-2}^3 g(x)\,dx where g(x) = x for \lvert x\rvert \ge 1 and g(x) = -x for \lvert x\rvert < 1, using G(x) = \tfrac12\lvert x^2 - 1\rvert; (c) show that \int_a^b \lvert x\rvert\,dx = B(b) - B(a) where B(x) = -\tfrac12 x^2 (x < 0), \tfrac12 x^2 (x \ge 0), stating the exceptional set E in (b) and (c). PYQ MAT-22 MCQ 27, 28, 29 · adapted also MAT-21 MCQ 19, 20 · adapted
15. Evaluate the integral \displaystyle\int_0^1 x^2(1 - \sqrt x)\,dx, justifying each hypothesis of the Fundamental Theorem of Calculus that you use. PYQ MAT-22 Q35
16. (a) If f is continuous on [-1,1], use the Substitution Theorem to show that \int_0^{\pi/2} f(\cos x)\,dx = \int_0^{\pi/2} f(\sin x)\,dx = \tfrac12\int_0^\pi f(\sin x)\,dx (and is not in general equal to \int_0^\pi f(\sin x)\,dx). (b) If f \in \mathcal{R}[a,b] and g(y) := f(y - c) on [a+c,b+c], show that \int_{a+c}^{b+c} g = \int_a^b f. PYQ MAT-22 MCQ 19, 26 · adapted
17. If f : [0,1] \to \mathbb{R} is continuous and \displaystyle\int_0^x f = \int_x^1 f for all x \in [0,1], prove that f \in \mathcal{R}[0,1] and f(x) = 0 for all x \in [0,1]. PYQ MAT-22 MCQ 30 · adapted
18. Let f be continuous on [a,b] with f(x) \ge 0 for

all x \in [a,b]. If \int_a^b f = 0, prove that f(x) = 0 for all x \in [a,b]. Show by an example that the continuity of f cannot be dropped. PYQ MAT-22 MCQ 23 · adapted
19. State the rule for differentiating G(x) = \int_{u(x)}^{v(x)} f(t)\,dt when f is continuous and u, v are differentiable. Hence find F'(x) on [0,1] for: (a) F(x) = \int_0^{x^2} (1 + t^3)^{-1}\,dt; (b) F(x) = \int_{x^2}^x \sqrt{1 + t^2}\,dt. PYQ MTS-22 Q14
20. (Mean Value Theorem for Integrals) If f : [a,b] \to \mathbb{R} is continuous on [a,b] (a < b), prove that there exists c \in [a,b] such that \int_a^b f = f(c)(b - a). Does this hold for step functions?
Section B · Paragraph / problem<small>All questions can be answered. Each carries 6 marks. Ceiling 72.</small>
21. Suppose f and g are in \mathcal{R}[a,b]. Prove that: (a) f + g \in \mathcal{R}[a,b] and \int_a^b (f + g) = \int_a^b f + \int_a^b g; (b) if f(x) \le g(x) for all x \in [a,b], then \int_a^b f \le \int_a^b g (and show why strict inequality < need not hold). PYQ MAT-21 Q29 also MTS-22 Q5, MTS-25 Q5 MAT-22 MCQ 14 · adapted
22. Define Thomae's function h on [0,1]. Prove directly from Definition 7.1.1 that h \in \mathcal{R}[0,1] and \int_0^1 h = 0. Explicitly list all elements of the set E = \{x \in [0,1] : h(x) \ge 1/5\}.
23. State and prove the Cauchy Criterion for the Riemann integrability of f : [a,b] \to \mathbb{R}. PYQ MAT-21 Q8
24. Prove using the Cauchy Criterion that Dirichlet's function (f(x) = 1 on \mathbb{Q} \cap [0,1], 0 on [0,1]\setminus\mathbb{Q}) is not Riemann integrable on [0,1]. Also prove that h(x) = x + 1 on \mathbb{Q} \cap [0,1], h(x) = 0 on [0,1]\setminus\mathbb{Q}, is not in \mathcal{R}[0,1]. PYQ MAT-22 Q4 also MAT-22 MCQ 17, MAT-21 MCQ 12, 16 · adapted
25. State and prove the Squeeze Theorem for Riemann integrals. Use it with step functions on an n-part equal partition to prove that h(x) = x belongs to \mathcal{R}[0,1] with \int_0^1 x\,dx = \tfrac12. PYQ MTS-26 Q18 also MTS-22 Q6, MAT-21 Q7
26. Prove that every step function \varphi : [a,b] \to \mathbb{R} is Riemann integrable, and derive the formula \int_a^b \varphi = \sum_{j=1}^m k_j(d_j - c_j). Use it to evaluate \int_0^3 g where g(x) = 2 on [0,1] and g(x) = 3 on (1,3]. PYQ MTS-26 Q5 also MAT-22 Q16
27. If f \in \mathcal{R}[a,b] and [c,d] \subseteq [a,b], prove that the restriction of f to [c,d] belongs to \mathcal{R}[c,d]. Deduce by mathematical induction that if a = c_0 < c_1 < \cdots < c_m = b, then \int_a^b f = \sum_{i=1}^m \int_{c_{i-1}}^{c_i} f. PYQ MAT-22 Q37b
28. Let f be bounded by M on [a,b] and suppose the restriction of f to [c,b] belongs to \mathcal{R}[c,b] for every c \in (a,b). Prove that f \in \mathcal{R}[a,b] and \lim_{c\to a^+} \int_c^b f = \int_a^b f. Deduce that g(x) = \sin(1/x) (x \in

(0,1]), g(0) = 0, belongs to \mathcal{R}[0,1], and show by an example that the boundedness of f cannot be dropped. PYQ MAT-22 MCQ 24, 25 · adapted also MAT-21 MCQ 18 · adapted
29. State and prove the First Form of the Fundamental Theorem of Calculus (prove the case E = \{a,b\} in full and show how the general finite set E follows). PYQ MTS-26 Q8
30. State and prove the Substitution Theorem for Riemann integrals. Use it to evaluate \int_1^4 \frac{\sin\sqrt t}{\sqrt t}\,dt, \int_1^4 \frac{\cos\sqrt t}{\sqrt t}\,dt, \int_0^1 t\sqrt{1+t^2}\,dt, \int_0^2 t^2(1+t^3)^{-1/2}\,dt, and \int_1^4 \frac{\sqrt{1+\sqrt t}}{\sqrt t}\,dt. Finally, explain why the Substitution Theorem with \varphi(t) = \sqrt t cannot be applied to \int_0^4 \frac{\sin\sqrt t}{\sqrt t}\,dt, and evaluate \int_0^4 \frac{\sin\sqrt t}{\sqrt t}\,dt using the First Form of the Fundamental Theorem. PYQ MTS-25 Q6 also MAT-22 Q18, MAT-21 Q19
31. State Lebesgue's Integrability Criterion. Use it to determine the Riemann integrability on [0,1] of: (a) any step function; (b) any monotone function; (c) G(1/n) = 1/n (n \in \mathbb{N}), G(x) = 0 elsewhere; (d) Dirichlet's function; (e) Thomae's function. PYQ MAT-22 Q6
32. State and prove the Composition Theorem for Riemann integrable functions. Show by an explicit counterexample that the continuity of the outer function \varphi cannot be dropped. PYQ MAT-21 Q4
33. If f \in \mathcal{R}[a,b] with \lvert f(x)\rvert \le M on [a,b], prove that \lvert f\rvert \in \mathcal{R}[a,b] and \left\lvert\int_a^b f\right\rvert \le \int_a^b \lvert f\rvert \le M(b - a). Show by an example that \lvert f\rvert \in \mathcal{R}[a,b] does not imply f \in \mathcal{R}[a,b]. PYQ MTS-25 Q8 also MTS-22 Q7, MAT-21 Q20
34. State and prove the Product Theorem: if f, g \in \mathcal{R}[a,b], then fg \in \mathcal{R}[a,b]. Give an example to show that \int_a^b fg \ne \left(\int_a^b f\right)\left(\int_a^b g\right) in general. PYQ MTS-26 Q6 also MAT-22 Q17, MAT-21 Q18
35. State and prove the Integration by Parts Theorem for Riemann integrals. Use it to evaluate \int_0^1 x e^x\,dx.
36. State and prove Taylor's Theorem with the Integral Form of the Remainder. Verify the theorem for f(x) = e^x on [0,1] (a = 0, b = 1) with n = 1 and n = 2. PYQ MTS-25 Q19 also MAT-22 Q32
Section C · Essay<small>Answer any two. Each carries 10 marks.</small>
37. Prove that if f : [a,b] \to \mathbb{R} is continuous on [a,b], then f \in \mathcal{R}[a,b]. State clearly where each hypothesis is used in the proof. Is every function in \mathcal{R}[a,b] continuous on [a,b]? Is every function in \mathcal{R}[a,b] uniformly continuous on [a,b]? Justify with examples. PYQ MTS-25 Q18 also MAT-21 Q30, MTS-22 Q19 MAT-22 MCQ 21, MAT-21 MCQ 15 ·

adapted
38. Prove that if f : [a,b] \to \mathbb{R} is monotone on [a,b], then f \in \mathcal{R}[a,b]. Illustrate the proof by computing the lower and upper step-function integrals \int_0^1 \alpha and \int_0^1 \omega for f(x) = x^2 on [0,1] with n = 4 equal subintervals. Is every function in \mathcal{R}[a,b] monotone on [a,b]? PYQ MTS-26 Q19 also MAT-22 Q30 MAT-22 MCQ 22 · adapted
39. State and prove the Additivity Theorem: let f : [a,b] \to \mathbb{R} and c \in (a,b); prove that f \in \mathcal{R}[a,b] if and only if its restrictions to [a,c] and [c,b] are both in \mathcal{R}, and in this case \int_a^b f = \int_a^c f + \int_c^b f. PYQ MTS-26 Q25 also MAT-22 Q37a, MTS-22 Q25
40. State and prove the Second Form of the Fundamental Theorem of Calculus (both the pointwise form at c \in [a,b] and the global consequence when f is continuous on [a,b]). Give an example of a function f \in \mathcal{R}[-1,1] whose indefinite integral fails to be differentiable at 0. PYQ MTS-25 Q25 also MAT-21 Q37a, MAT-22 Q33
<section class="page"> <h2>Coverage</h2> <table class="t">
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31 MCQs).</p> </section>
=== NOTES / DOUBTS (STAGE 4 OF 6) ===

- Completed Stage 4 (src/m3/paper.html): all 40 questions (Section A Q1–Q20 × 3 marks, Section B Q21–Q36 × 6 marks, Section C Q37–Q40 × 10 marks) following brief §6 (content_m3.md) and Part 8 (pyq_all.md), with the Coverage table and the closing line confirming all 74 past questions (and all 77 rows in Part 8) appear.
- Next up in Stage 5: src/m3/answers_1.html and src/m3/answers_2.html (Formal solutions to Q1–Q28).
- Reminder for Stage 6: Please paste or attach figs.py and figs_m2.py before or when requesting Stage 6 so src/m3/figs_m3.py can be generated using the exact S and Plot classes.